



Pre installation

Installation

Operating

**LC 18MT/WC Chilled Water System for
GE Signa *TwinSpeed* MRI System**

**Technical Publication 471
Revision 1**

The People of Ellis and Watts are committed to set goals and objectives to meet the expectations and needs of our customers. Responsibility and ownership at the point of operation will enhance the success of our mission

Ellis and Watts International, Inc.
Batavia, Ohio 45103

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REVISION HISTORY PAGE

REVISION NUMBER	DATE	REASON FOR CHANGE	AUTHOR
PRELIMINARY	05/23/01	NEW PRODUCT RELEASE	CAR
A	7/20/01	REVISIONS PER GE	CAR
B	7/30/01	REVISIONS PER GE	CAR
C	8/13/01	REVISIONS PER GE	CAR
D	10/26/01	REVISIONS PER GE	CAR
E	11/21/01	REVISE PAGE 15	CAR
0	11/1/01	INITIAL RELEASE	CAR
1	11/29/01	UPDATE PAGES 11, 15, 55	CAR

1 INTRODUCTION

1-1 CONTACT ELLIS & WATTS

If a non-service issue arises concerning the equipment or the documentation provided with the chiller you can contact Ellis & Watts in writing at the following address, by phone, or via E-mail at the Ellis & Watts web page.

**Ellis & Watts, International, Inc.
Thermal Sales
4400 Glen Willow Lake Lane
Batavia, OH 45103**

Phone: 513-752-9000

Internet: WWW.ELLISWATTS.COM

Ellis & Watts provides 24-hour service support in the event that the Ellis and Watts equipment fails to perform properly. Service calls are processed more easily if you provide the service support department with the chiller **model number, serial number, the site name and location, and a description of the problem including the alarm or fault displayed on the operator interface**. The service support department will dispatch a qualified local service technician to correct the problem.

The 24 hour service support department phone numbers are:

**Domestic Toll Free: 1-888-744-3195
International: 513-768-3195**

1-2 WARRANTY REPLACEMENT PART PROCEDURE

To initiate a warranty claim with Ellis & Watts, contact the Thermal Sales Department. The equipment model number and serial number are required to process a warranty claim. The Thermal Sales representative will issue a Return Goods Authorization (RGA) number. The RGA number must be on the returned part's package.

NOTICE

Items returned without an RGA number will not be processed until the necessary information is obtained from the sender.

1-3 INSTALLATION TOOLS & EQUIPMENT

Item Description	QTY	Used For	Supplied By
10 foot (3.05 m) Stepladder	2	Uncrating Equipment	Installer
Forklift, 2 ton (1814 kg) Minimum	1	Lifting and Moving TSCC	Installer
Drill 3/8 inch minimum	1	Drill mounting holes	Installer
Eye, Lift Straps 2 inch W x 18ft. Lg. (5.08cm W x 5.49m Lg.)	2	Lifting TSCC off Shipping Skid	Installer
C Clamp, 10 inch (25.4cm)	2	Securing Straps	Installer
Pry bar, lever, 5000 lb. (2250kg)	1	Lifting Equipment	Installer
4X4X4 inch wood block (10.16X10.16X10.16 cm)	4	Supporting TSCC Cabinet, Hose Assembling	Installer
Wood Block Approximately 1X4X6 inch (2.54 x 10.16 x 15.24cm)	1	Supporting TSCC Cabinet	Installer
Metal Snips	1	Cut banding Straps	Installer
4 Conductor (4AWG/25mm ²) power cable minimum	1	Power for TSCC Cabinet	Installer
3/8 inch (M10) Hardware Length to be determined	8	For tie down bracket	Installer
Light oil or soapy water	1 oz	Lubricating Hose Barbs	Installer
LocTite 567 or equivalent	4 oz	Sealing Pipe Threads	Installer
Razor Knife	1	Cutting Hose	Installer
Crowbar, 2 foot (.61m)	1	Uncrating Equipment	Installer
1 ½ inch flexible lines and 1 ½" male NPT connections	2	Piping condenser water supply and return	Installer

Table 1 **Installation Tools And Supplies**

1-4-1 Installation Responsibility Matrix

	Cust. Contr.	GE Service	E&W Service	Who Pays
Once Cooling Cabinets Arrive				
Unload TSCC from Truck	X			Customer
Move TSCC to Equipment Room and mount (Use seismic Tie down brackets where required by law). See Section 3-5, 6	X			Customer
Connect customer supplied power cable from MDP to TSCC. See Section 5-1-1	X			Customer
Install condenser water supply and return lines for water cooled TSCC . See Section 3-2-3	X			Customer

	Cust. Contr.	GE Service	E&W Service	Who Pays
Once Magnet Arrives				
Install water lines from TSCC to 1) Gradient Coil 2) Cryo Cooler Compressor (Note: If Gradient is not connected, then it must be short looped on TSCC). See Section 5-1		X		GE
Fill TSCC with Propylene Glycol. See Section 6-1		X		GE
Start up TSCC and ensure proper phasing and no coolant leaks. See Section 6-1		X		GE
Verify proper operation per E&W Chiller Startup Check List after Cryo Cooler Compressor and Gradient Coil load is present			X	E&W

Table 2 **Installation Responsibility Matrix**

1-4 EXPLANATION OF NOTES, CAUTIONS, AND WARNINGS

A **NOTE** is used to provide supplemental information or direction.



CAUTION

A **CAUTION** is used to call particular attention to a step of a procedure, which, if not strictly followed, could result in damage to, or destruction of equipment.



WARNING

A **WARNING** is used to call attention to a particular step of a procedure, which, if not strictly followed, could result in serious injury, or death of personnel.

1-5 SAFETY WARNINGS**WARNING**

NO PERSON SHALL REACH INTO THE UNIT ENCLOSURE, WHICH CONTAINS LIVE ELECTRICAL CIRCUITS AS WELL AS ROTATING PARTS, FOR THE PURPOSE OF INSPECTING, SERVICING, ADJUSTING, OR MAINTENANCE EXCEPT IN THE PRESENCE OF SOMEONE CAPABLE OF RENDERING AID. CONTACTING SUCH ITEMS COULD CAUSE SERIOUS BODILY INJURY AND OR DEATH.

WARNING

PERSONNEL WORKING NEAR HIGH VOLTAGES SHOULD BE IN THE PRESENCE OF SOMEONE HAVING RESUSCITATION SKILLS. COMING IN CONTACT WITH HIGH VOLTAGE COULD CAUSE SERIOUS BODILY INJURY AND OR DEATH.

WARNING

TURN OFF AND TAG OUT THE MAIN POWER AT THE DISCONNECT, DURING ANY SERVICING OR MAINTENANCE OPERATIONS. REMOTE STARTING COMPONENTS OR DIRECT CONTACT WITH ELECTRICAL COMPONENTS MAY RESULT IN INJURY TO PERSONNEL.

WARNING

TURN OFF AND TAG OUT THE DISCONNECT TO THE UNIT BEFORE WORKING ON CONTACTORS OR OTHER COMPONENTS IN THE ELECTRIC BOX. WHEN THE MAIN POWER SUPPLY TO THE UNIT IS ON, THE LIVE SIDE OF EACH CONTACTOR CONTAINS HIGH VOLTAGE REGARDLESS OF ANY OTHER SWITCH SETTING. DO NOT ATTEMPT MAINTENANCE PROCEDURES INSIDE EQUIPMENT WITH HIGH VOLTAGE ON. REMOVE POWER AND TAG OUT THE DISCONNECT. GROUND ELECTRICAL COMPONENTS PRIOR TO TOUCHING TO PREVENT BODILY INJURY.

WARNING

NO SMOKING OR OPEN FLAMES WITHIN 50 feet (15 M) OF REFRIGERANT SYSTEM. REFRIGERANT COMING IN CONTACT WITH LIT CIGARETTES CAN CREATE AN IRRITATING AND POSSIBLY TOXIC GAS.

SAFETY WARNINGS CONTINUED**WARNING**

STAND CLEAR DURING LIFTING OPERATIONS. OPERATOR ERROR AND/OR EQUIPMENT MALFUNCTION CAN CREATE OPPORTUNITIES FOR PERSONAL INJURY.

WARNING

THE UNIT HAS A CLOSED LOOP HIGH PRESSURE REFRIGERANT SYSTEM. DO NOT CLOSE, OPEN OR ADJUST VALVES IN THE REFRIGERANT SYSTEM UNLESS A SPECIFIC INSTRUCTION STATES THAT VALVES MUST BE OPENED, CLOSED OR ADJUSTED. PERSONNEL COULD BE INJURED OR EQUIPMENT COULD BE DAMAGED.

WARNING

BEFORE BRAZING OR SOLDERING ANY REFRIGERATION COMPONENT, RECOVER REFRIGERANT PER US ENVIRONMENTAL PROTECTION AGENCY (EPA) APPROVED PROCEDURE TO RELEASE THE PRESSURE FROM THE REFRIGERATION SYSTEM. SUDDEN DECOMPRESSION OF REFRIGERANT PIPING CAN CAUSE SERIOUS PERSONAL INJURY.

WARNING

USE EXTREME CARE WHEN BRAZING OR SOLDERING IN THE PRESENCE OF R-134a. WHEN R-134a COMES IN CONTACT WITH HOT METAL, AN IRRITATING GAS CAN FORM. HAVE SUFFICIENT FRESH AIR AND VENTILATION WHEN BRAZING, SOLDERING OR USING A HALIDE TORCH LEAK DETECTOR.

WARNING

THE COMPRESSOR MUST ALWAYS BE OPERATED WITH THE COMPRESSOR DISCHARGE SERVICE VALVE IN THE BACK-SEAT (FULLY OPEN) POSITION. FAILURE TO DO SO CAN CAUSE SERIOUS COMPONENT FAILURE.

SAFETY WARNINGS CONTINUED

WARNING

WHEN AN ABNORMAL CONDITION IS NOTED AND PROCEDURES IN THIS MANUAL DO NOT SPECIFICALLY RELATE TO THE NOTED CONDITION, WORK SHALL BE STOPPED. TECHNICALLY QUALIFIED PERSONNEL SHALL GIVE GUIDANCE ON THE ABNORMAL CONDITION BEFORE WORK CAN BE CONTINUED, TO PREVENT POSSIBLE PERSONAL INJURY OR EQUIPMENT DAMAGE.

2 SHIPPING AND DELIVERY DATA

The General Electric Medical Systems (GEMS) part number 2281346 High Voltage Indoor Water Cooled Unit is the Ellis and Watts chiller model LC18MT/WC. The LC18MT/WC consists of a liquid chiller (TSCC).

2-1 SHIPMENT

For domestic shipment, the TSCC will be shipped to the construction site. International shipment shall be shipped to a collection point where all components for the magnet will be collected and shipped to the construction site.

	Height	Width	Depth	Weights
TSCC	82 inches (208.28 cm)	58 inches (147.32 cm)	37.5 inches (95.25 cm)	1,100 lbs. (499 kg)
Ship Loose Box		28.5 inches (72.5 cm)	58.8 inches (149.4 cm)	140 lbs. (63.5 kg)

Table 3 **Shipping Dimensions**

The ship loose items are boxed and shipped together on a wooden skid. Domestic shipments are covered with cardboard and banded on wooden skids. International shipments are domestic packages shipped in crates.

2-2 STORAGE REQUIREMENTS

The non-operation environmental conditions are as follows:

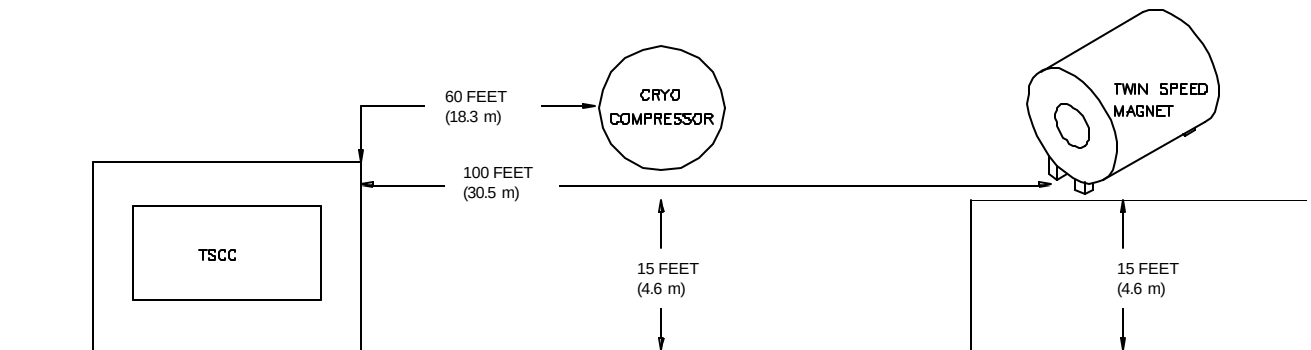
ITEM	REQUIREMENT
Ambient Temperature	-40°C to 70°C
Humidity	5% to 100% RH
Altitude	984 feet (300 m) below sea level 17,060 feet (5,200 m) above sea level
Acceleration	+/- 1.5 G any axis
Magnetic Field	30 Gauss
NOTE: Do not expose to water (rain, snow)	

Table 4 **Environmental Conditions**

NOTE: The TSCC should be kept in its shipping crate until it has been moved to the installation site. When uncrating, care should be taken to prevent damage.

3 PRE-INSTALLATION REQUIREMENTS

3-1 EQUIPMENT PLACEMENT



The maximum distance from the TSCC to the magnet is 15 feet (4.6 m) vertical and 100 feet (30.5 m) total. The maximum distance from the TSCC to the Cryo Cooler is 15 feet (4.6 m) vertical and 60 feet (18.3 m) total.

Illustration 1 **PLACEMENT OF TSCC TO MAGNET**

3-2 SITE ENVIRONMENT

3-2-1 Operating Specifications

ITEM	REQUIREMENT
Ambient Temperature	5°C to 40°C
Humidity	10% to 90% RH
Altitude	984 feet (300 m) below sea level 11,811 feet (3,600 m) above sea level
Magnetic Field	30 Gauss
Noise Levels	75 dBa @ 1 meter in all directions

Table 5 **Operating Specifications**

3-2-2 Power Requirements

ITEM	REQUIREMENT
Wire – <u>NOTE</u>	6 AWG (minimum) 3AWG (maximum) 3 conductor w/ground
Power Consumption	14 KVA (maximum)
Voltage Requirements	480 VAC $\pm$ 10% 60 Hz $\pm$ 3 Hz 400 VAC $\pm$ 10% 50 Hz $\pm$ 3 Hz
Circuit Size	50 AMP (maximum)
<u>NOTE:</u> The TSCC will be hardwired to the MDP (Main Distribution Panel) with a four conductor power cable supplied by the installer. Wire size determined by the installer.	

Table 6 **Power Requirements**

3-2-3 Condenser Water Cooling Requirements

ITEM	REQUIREMENT
pH	6.5 – 8.2
Hardness	Less than 200 PPM of Calcium Carbonate
Suspended Matter	Less than 10 mg/L, less than 150 micron particle size
Anti-Freeze	Propylene Glycol (50% maximum)
Minimum and Maximum Pressure	25 psi – 150 psi
Pressure Drop	96.5 kPa @ 47 L/min; 24.1 kPa @ 33 L/min
Maximum Heat Output to Water	100,000 BTU/Hr
Maximum Heat Output to Room	12,000 BTU/Hr

Table 7 Customer Water Cooling Requirements

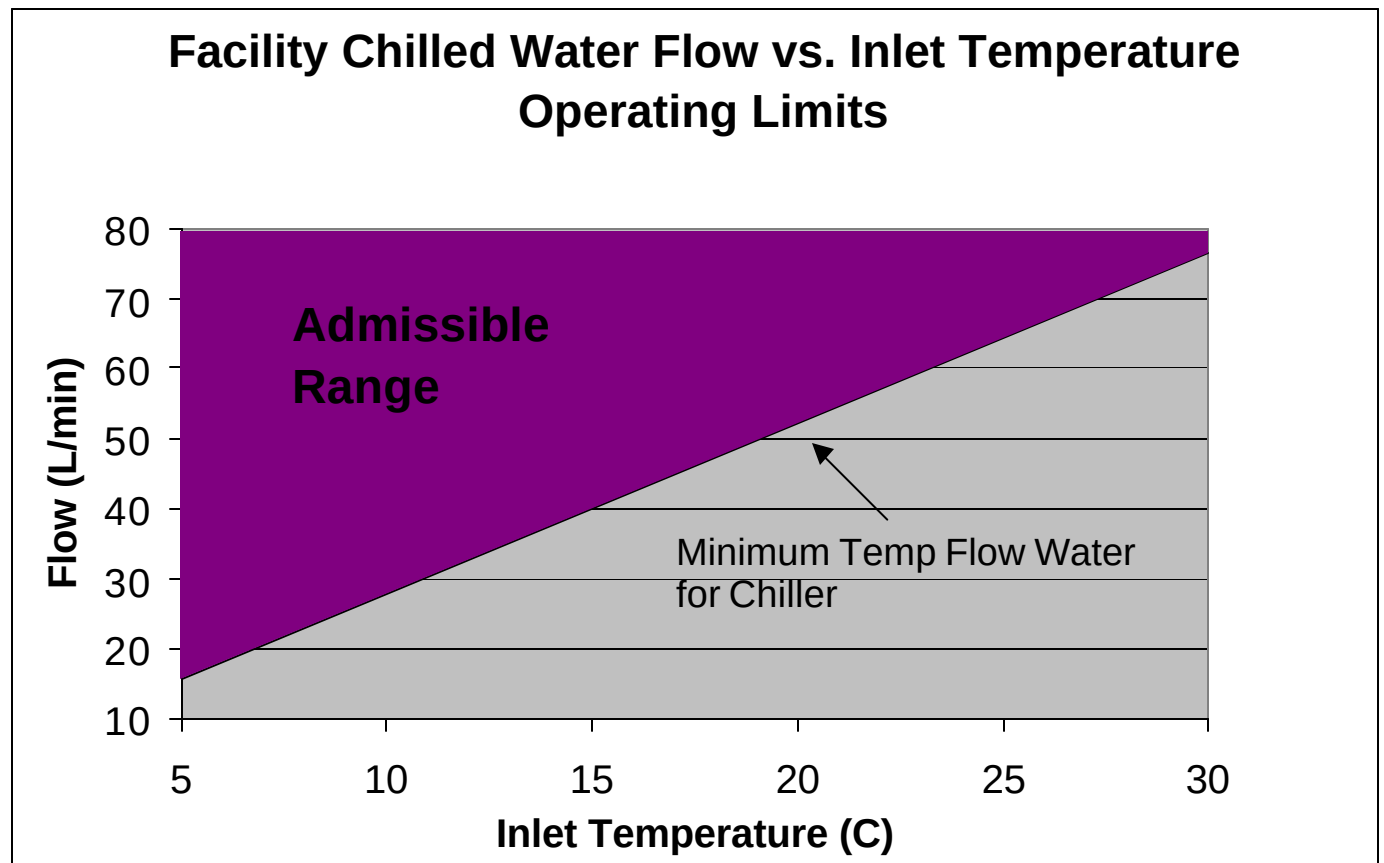


Illustration 2 FLOW AND TEMPERATURE LIMITS FOR COOLING WATER

3-3 EQUIPMENT CLEARANCE REQUIREMENTS

It is necessary to maintain proper distances at the front and back of the TSCC cabinet in order to have a good service area. There is no requirement for clearance at the side of the cabinet, but any allowances would aid in the service of the TSCC cabinet.

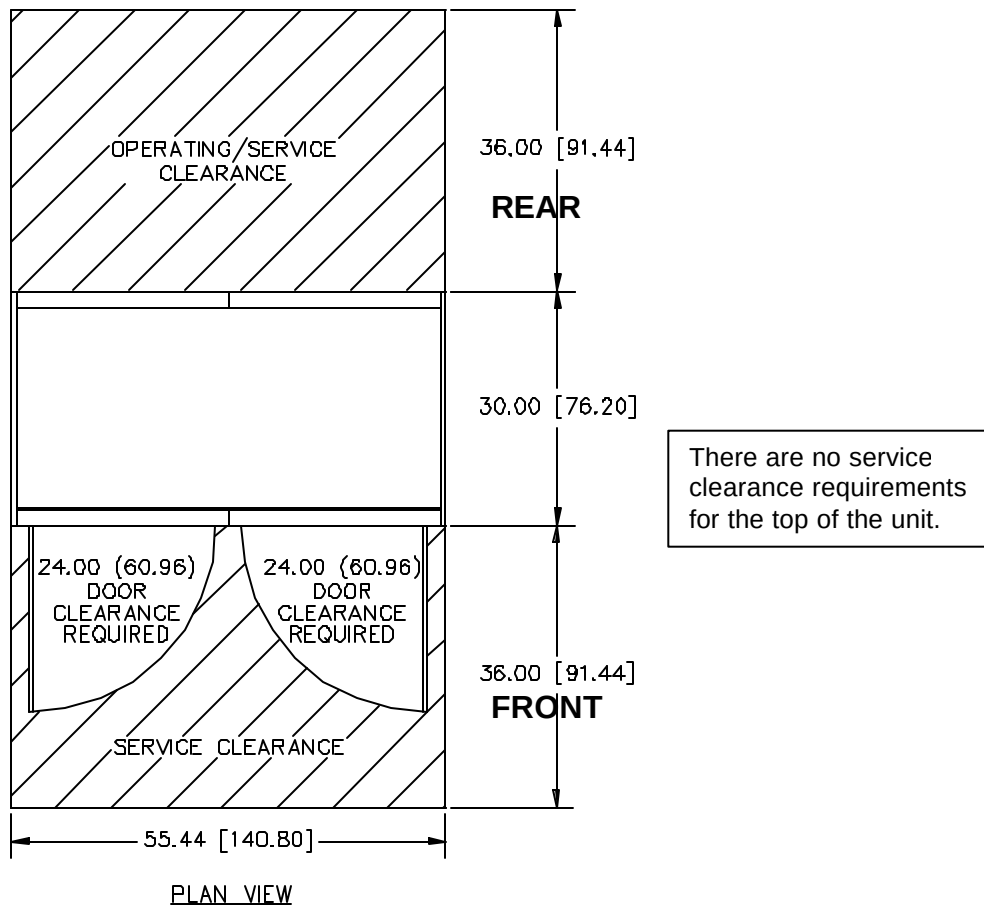


Illustration 3 **NECESSARY CLEARANCES FOR TSCC**

3-4 COMPONENT DIMENSIONS

Uncrated Dimensions

Component	Hallway / Door Height	Hallway Width <u>NOTE</u>	Doorway Depth	Installed Weight
TSCC	73 inches (185 cm)	55.44 inches (140.8cm)	33.59 inches (85.32 cm)	1,000 pounds (454 kg)
NOTE: minimum width for 90° turns				

Table 8 **Uncrated Dimensions**

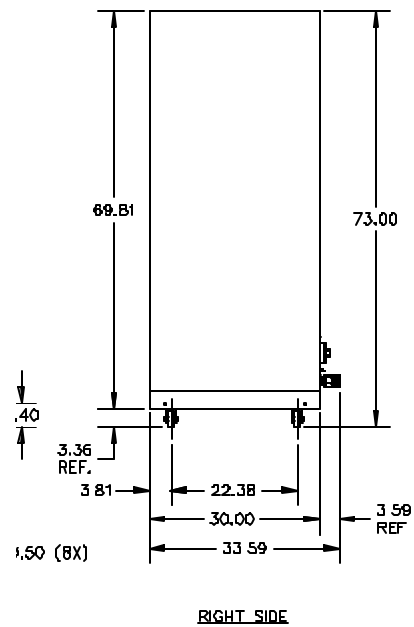
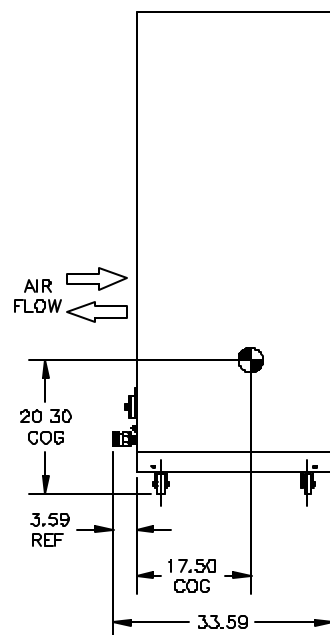
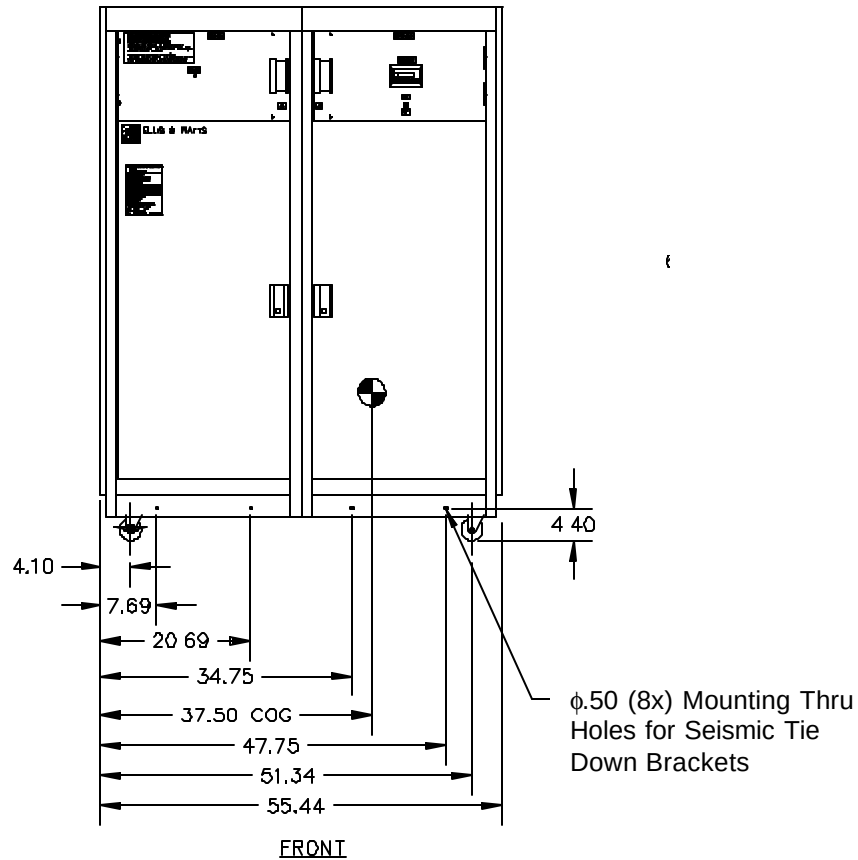
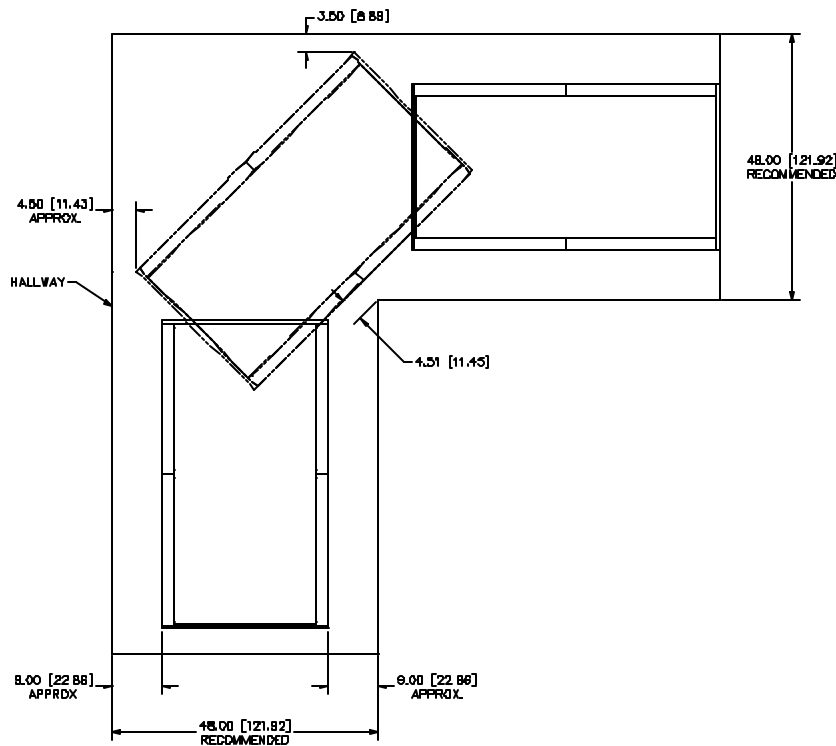
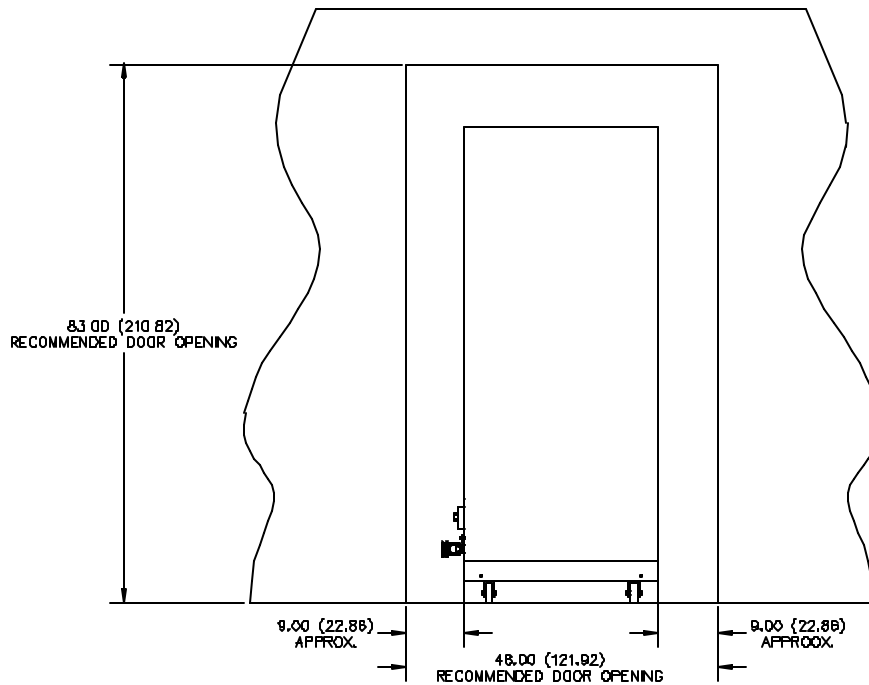


Illustration 4 TSCC OVERALL DIMENSIONS

3-5 MINIMUM DOOR/HALLWAY SIZES

Component	Hallway / Door Height	Hallway Width *NOTE	Standard Doorway Width
TSCC	83 inches 210.82 cm	48 inches 121.92 cm	40 inches 101.6 cm
NOTE: Minimum width for 90° turns			

Table 9 **Door/Hallway Sizes****Illustration 5 MINIMUM HALLWAY DIMENSION FOR TURNS**

Illustration 6 **MINIMUM DOORWAY DIMENSIONS****3-6 DOORWAY THRESHOLD CONSIDERATION**

The clearance from the bottom of the TSCC cabinet to the floor is approximately 3.36 inches (8.53 cm) therefore doorway threshold should be as smooth as possible or special considerations have to be made to move and roll the cabinet over a threshold height of 3 inches (7.62 cm)

3-7 FLOOR LOADING AND WEIGHTS

EQUIPMENT	INSTALLED WEIGHT
TSCC	1000 lbs. (454 Kg)

Table 10 **Floor Loading Requirements**

3-8 ANCHORING AND SEISMIC CONSIDERATIONS

The TSCC cabinet has tie down brackets included in the ship loose parts, which should be used to aid in preventing the cabinet from shifting during a seismic event. Site Architects and Engineers need to take into consideration the raised floor-framing layout in order to have substantial support for the tie down brackets.

Sites requiring seismic anchoring by code should have the site architect and engineer review the response spectra and/or United Builders Code (UBC) for their location and then contact Ellis and Watts International (Call 513-752-9000) for assistance in seismic planning of the site.

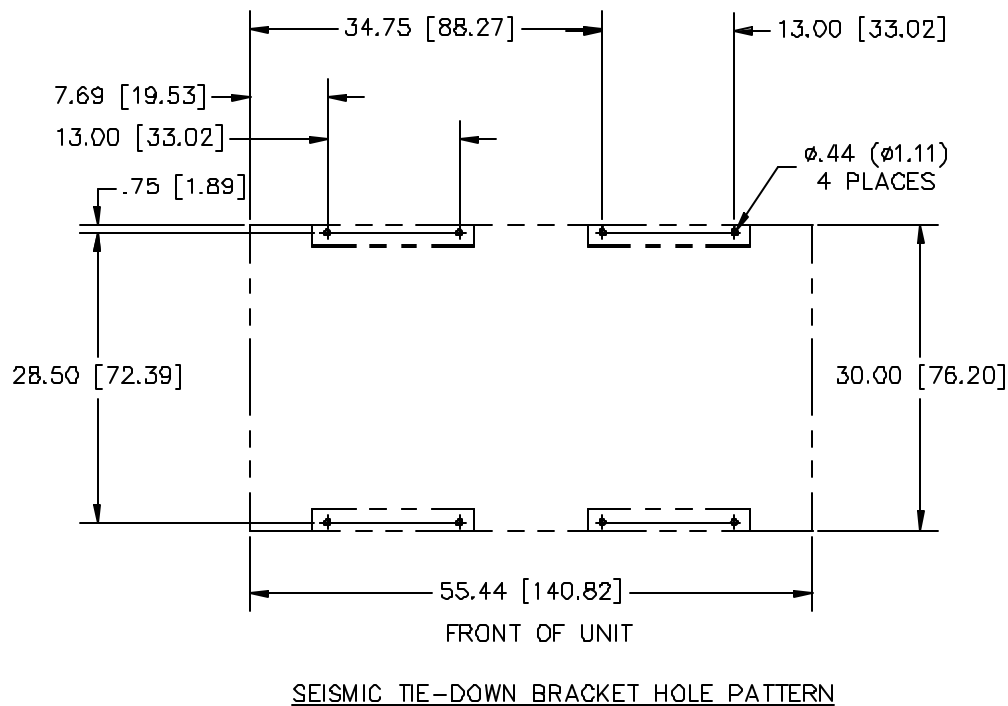


Illustration 7 TSCC TIE DOWN BRACKET PLACEMENT

4 UNCRATING AND INSTALLATION PROCEDURES

4-1 UNCRATING AND RECEIVING INSPECTION

NOTE

Steps 1 through 3 are for International shipments only.

4-1-1 TSCC

1. Using a #2 Phillips screwdriver, unscrew the wood screws holding on the top of the crate.

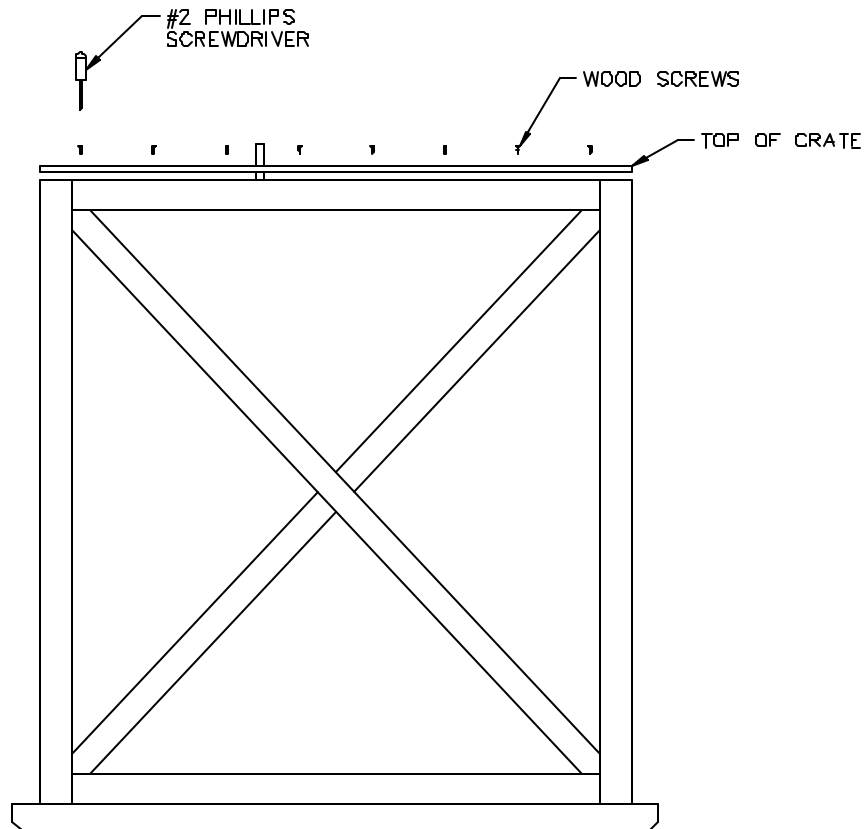


Illustration 8 INTERNATIONAL CRATE EXAMPLE

2. Remove top of the crate and place in an area clear of the uncrating area.
3. Using a pry bar carefully remove the front, sides, and back of the crate and place in an area clear of the uncrating area.

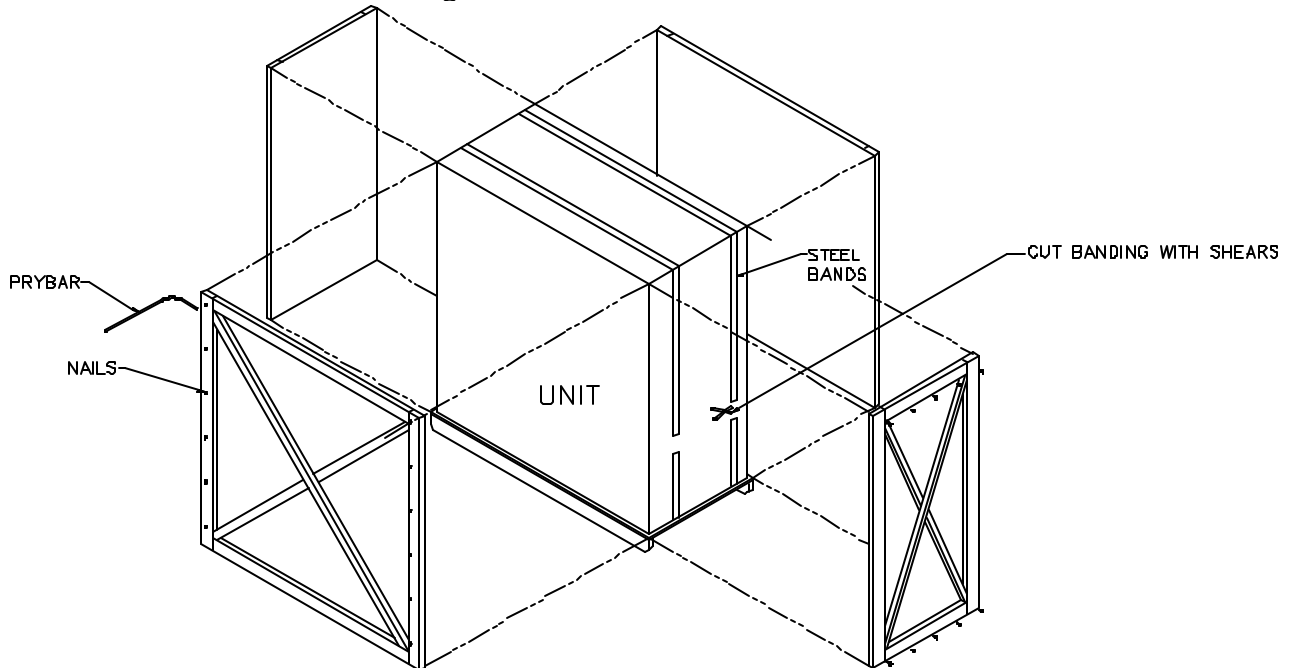


Illustration 9 **UNCRATING**

CAUTION

Banding straps can be under high tension and can whip out when cut.

4. Remove banding by cutting with metal cutting hand shears.
5. CAREFULLY remove plastic.
6. Inspect equipment for any damage. Contact Ellis and Watts and the shipping company as soon as possible and report any discovered damage.

NOTE: There are seven boxes that come with the TSCC that should be placed in a secure location for use during installation. These boxes contain the ship loose items and must be stored in a safe location for later use. Boxes can be inventoried using the list in section 4-2.

7. Using two eye lifting straps approximately 2 inches (5.08 cm) wide and 18 feet (5.49 m) long, create a basket lift by pushing one end under unit (between board slats from front to back), and up opposite side until lifting strap eyes are even.

8. Do this in two places, evenly spaced from, and as near as possible to sides of TSCC as fork spread will allow.
9. Using two 10 foot (3.05 m) stepladders on each end of the unit, hold up straps while bringing in forklift as near as possible to front or back of TSCC cabinet so lift forks can be inserted into strap eyes.

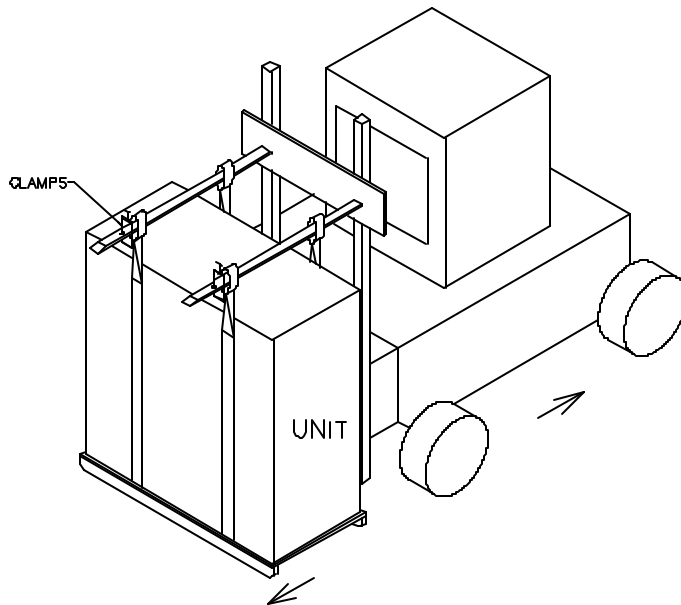


Illustration 10 TSCC LIFTING METHOD

10. After sliding lift forks through strap eyes, attach a C-clamp or similar clamping device, onto the end of the forks to prevent straps from sliding off the forks

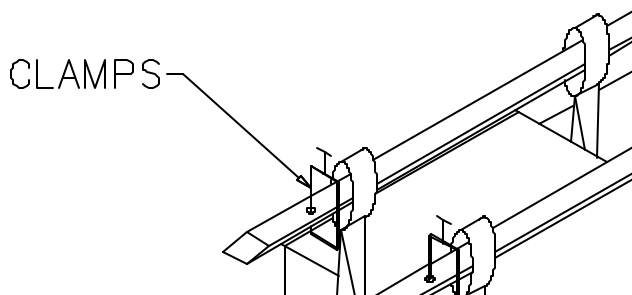


Illustration 11 C-CLAMP PLACEMENT

11. Gently lift TSCC cabinet until clear of shipping skid.

12. Remove skid, carefully move unit to smooth level area where TSCC cabinet can be moved on attached casters.

13. Remove slings from lifting forks and slide out from under TSCC cabinet.

The previous steps define the preferred method of removing the TSCC cabinet from the skid. Other methods of removing the TSCC cabinet from the shipping skid may be acceptable but could involve many more detailed steps. It is up to the judgment of the riggers and movers to determine the specific method of handling the TSCC cabinet.

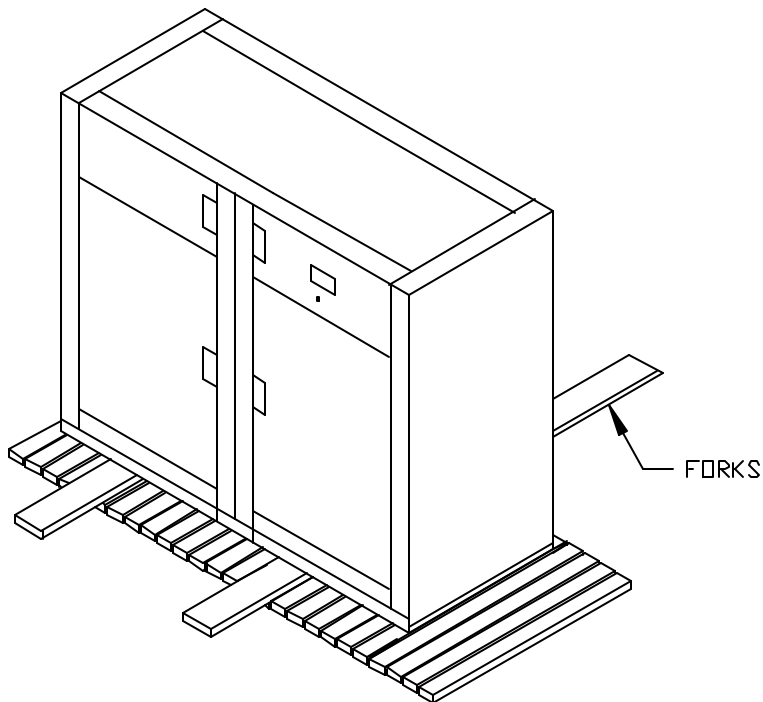


Illustration 12 **ACCEPTABLE ALTERNATIVE LIFTING METHOD**

4-2 SHIP LOOSE ITEMS

The following tables list the ship loose items that are necessary for the proper installation of the Ellis and Watts equipment. The following tables can be used to verify the inventory of the shipped loose items. Also, the field installation drawings FI-0023A, FI-0024A, FI-0026A, FI-0058A are included in the ship loose kits.

The TSCC is shipped with the following field installation kits.

No.	Part No.	Ship Loose Items Description	QTY EA
1	T2624-0201-21	Seismic Tie Down Bracket	4
2	50-19X-633	Screw, Truss Head Philips, ¼ -20 x 1 ¼ inch LG SST	8
3	50-3E-34	Washer, Flat, ¼ inch SST	8
4	50-2Z-126	Nut, Lock, ¼ - 20 SST	8
5	FI-0023A	Installation Drawing	1

Table 11 **Seismic Tie Down Install Kit**

No.	Part No.	Ship Loose Items Description	QTY EA
1	T2637-140ZS	¾ inch Q.D. Zinc Plated Steel Nipple, NPT Male	1
2	T2637-142ZS	¾ inch Q.D. Zinc Plated Steel Coupler, NPT Female	1
3	T2590-111ZS	1/2 inch Q.D. Zinc Plated Steel Nipple, NPT Male	1
4	T2590-113ZS	1/2 inch Q.D. Zinc Plated Steel Coupler, NPT Female	1
5	J1476-300ZS	¾ inch Hose Barb, ¾ inch NPT Male	2
6	J1476-301ZS	½ inch Hose Barb, ½ inch NPT Male	2
7	SS1045-08	Tube, ½ inch, Push-Lok 120 feet	1
8	FI-0024A	Drawing, Piping/Hose Connections	1

Table 12 **Piping/Hosing Connections Install Kit**

NO.	PART NO.	SHIPS LOOSE ITEMS DISCRIPTION	QTY EA
1	FI-0026	Drawing, Electrical/Power/Communications Connections	1

Table 13 **Electrical/Power/Communications Install Kit**

NO.	PART NO.	SHIPS LOOSE ITEMS DISCRIPTION	QTY EA
1	T2625-279A	Filter, 10 inch Cartridge	1
2	SS627-7-A20	Fuse, Time Delay, 1.0 Amp, 500V	2
3	E & W P/N J1520-5050	Propylene Glycol 1 Gal. GE P/N 2297672	12
4	FI-0058A	Drawing, Spare Parts	1

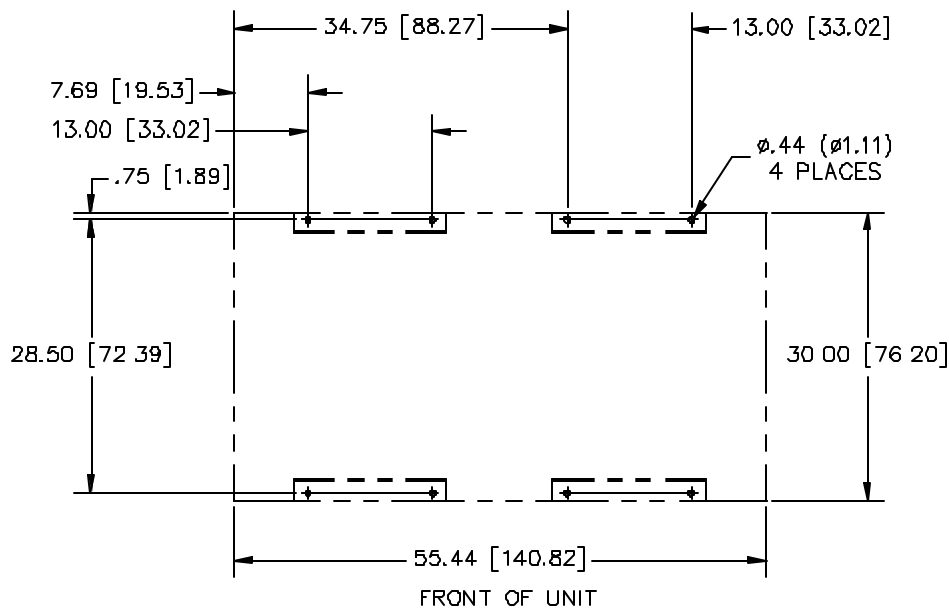
Table 14 **Spare Parts Install Kit**

5 EQUIPMENT INSTALLATION

5-1 TIE DOWN BRACKET LAYOUT PATTERN

Refer to drawing FI-0023A that comes with the ship loose parts, for detailed illustrations.

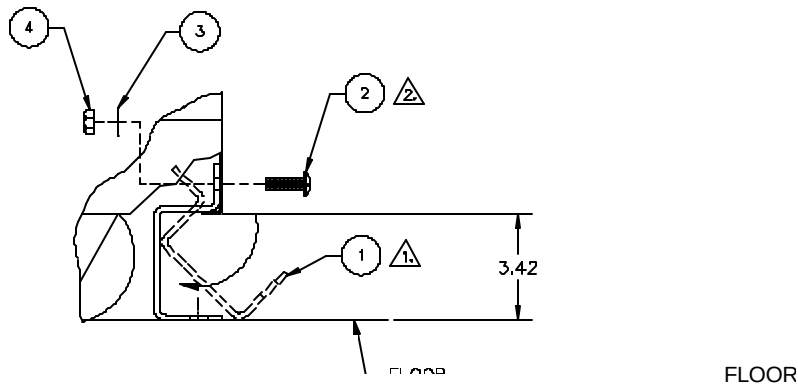
1. Locate-Tie Down Brackets in the Seismic Tie Down Install Kit.
2. Locate positions to mount brackets per Tie-Down Brackets Layout (see Illustration 14).



SEISMIC TIE-DOWN BRACKET HOLE PATTERN

Illustration 13 TIE-DOWN BRACKET LAYOUT PATTERN

3. Locate mounting hardware in Seismic Tie Down Install Kit.
Refer to Table 11, items 2-4.
4. Take one $\frac{1}{2}$ x 4 inch long bolt, slide one $\frac{1}{2}$ inch washer over bolt and then push the bolt through the hole in the base frame from the bottom up. Slide one $\frac{1}{2}$ inch washer over bolt and thread $\frac{1}{2}$ inch lock nut down until it is snug against the frame member and tighten approximately $\frac{1}{4}$ to $\frac{1}{2}$ turn more. See Illustration 14.

**Illustration 14 TIE DOWN MOUNTING BRACKET**

Refer to table 11, item number 1 to 4

5. Place the tie down bracket over the $\frac{1}{2}$ inch bolt (six per unit) at the bottom of the frame. Use the tie down bracket as a template and mark the location for the floor mounting hardware.
6. Repeat item 3 for the other 3 brackets.
7. Remove brackets and drill marked locations to the appropriate size for installer supplied hardware. Ellis and Watts recommend using $\frac{3}{8}$ inch hardware.
8. Place the tie down bracket over the $\frac{1}{2}$ inch bolt at the bottom of the frame, Locate one $\frac{1}{2}$ inch lock-nut and one $\frac{1}{2}$ inch washer. Slide washer down the bolt onto the tie-down bracket and thread the lock-nut down until it stops and then tighten approximately one more turn.

5-1-1 Electrical Installation

Power Wiring

Connecting power to the TSCC should only be performed by trained electrical installation personnel that have a good knowledge of lock-out/tag-out procedures. Short circuit and field wiring must be accomplished in accordance to the National Electrical code, ANSI/DEPA No. 70-1990 and or the Canadian Electrical Code, Part 1, CSA Standard C22.1-1990 and local laws and regulations as they exist.

Power Requirement

480 V +/- 10%, 60 Hz +/- 3 Hz, 3 phase, WYE or Delta

or

400 V +/- 10%, 50 Hz +/- 3 Hz, 3 Phase, WYE or Delta

Full Load current is 20 kVA.

Locate the disconnect switch on the back lower right side of the TSCC.

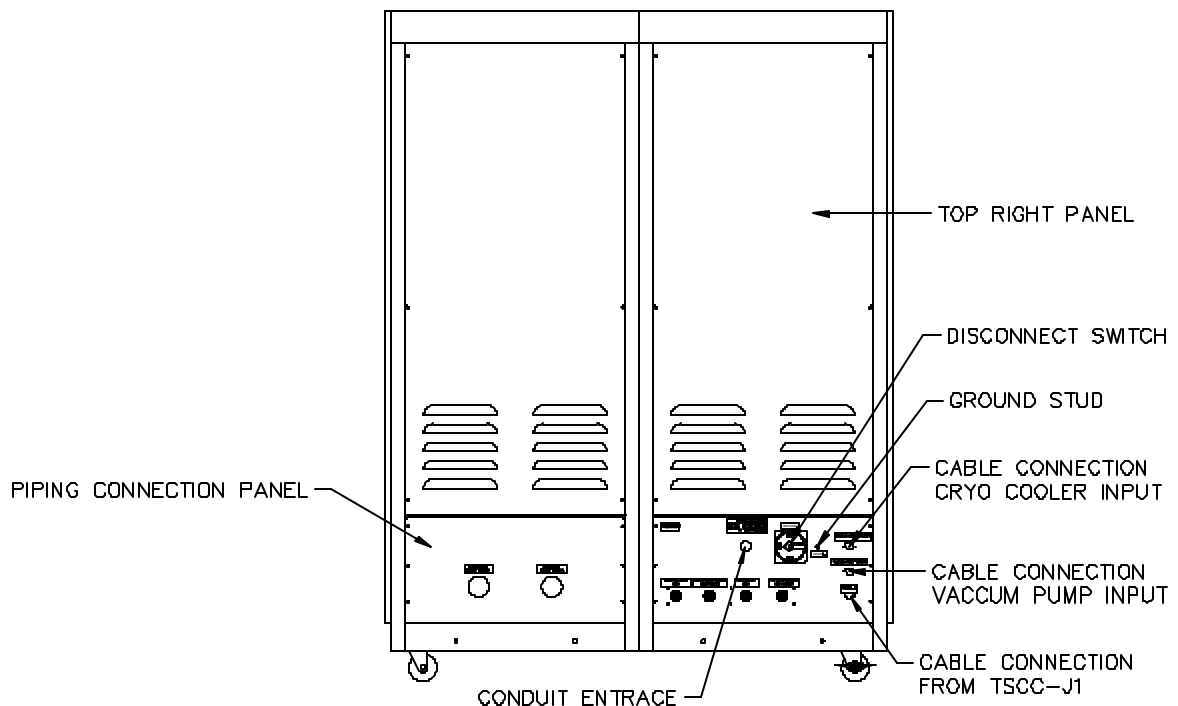


Illustration 15 **JUNCTION BOX LOCATION**

1. Remove the top right panel cover by loosening the six cover screws and sliding the panel off.

NOTE: Ellis & Watts recommends using a minimum conductor size of 6 AWG. Maximum size recommended is 3 AWG. Ellis & Watts recommends using 1 inch (minimum) Liquidtite (or equivalent) conduit and fittings. The installer supplies wire, conduit, and fittings.

2. Remove the guard protectors from the back of the switch (See Illustration 17).
3. Route the four power conductors through the 1-1/8 inch hole provided on the panel.
4. Strip the wire and insert into the terminals. When the conductor is properly installed, there should be no bare wire exposed. Pay special attention to the phase rotation.

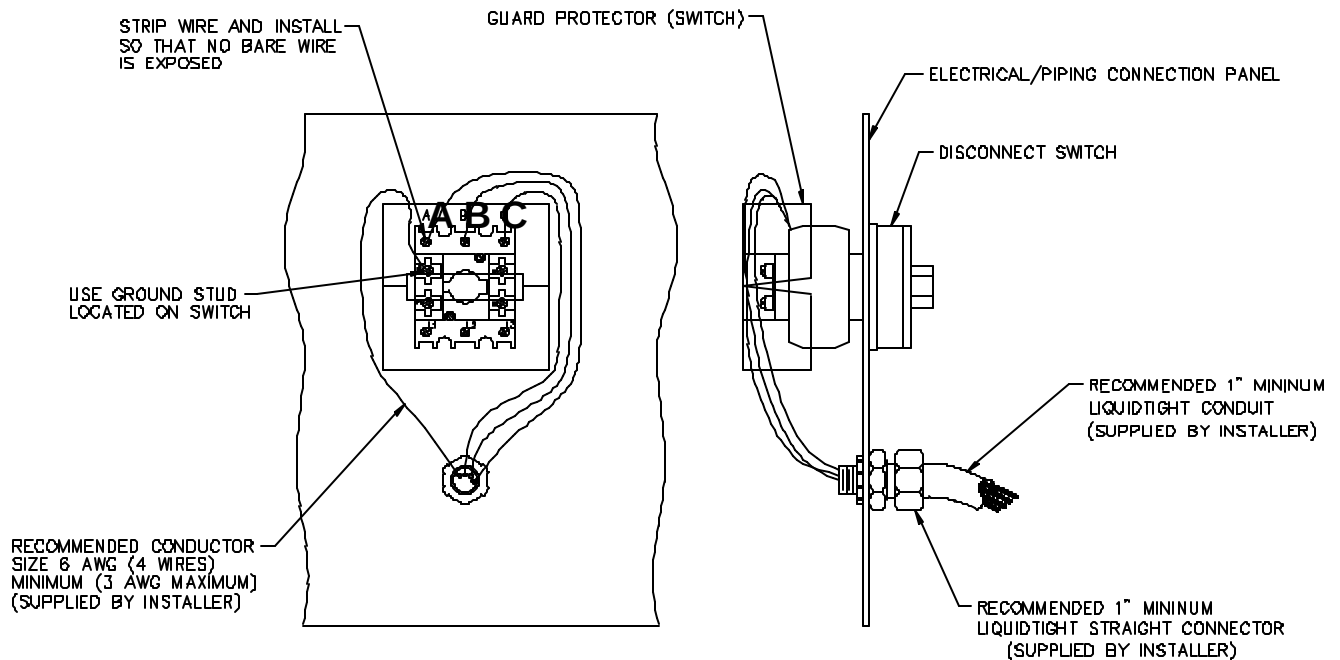


Illustration 16 **JUNCTION BOX WIRING**

5. Attach the ground wire to the ground stud provided on the terminal.
6. Reinstall the guard protectors and panel cover.

5-1-2 Assembling The Interconnecting Hoses

Verify the following parts and quantities from the Piping/Hosing Install Kit.

Refer to drawing FI-0024A that comes with the ship loose parts, for detailed instructions.

The following supplies are supplied by the installer and are required for the proper installation of the interconnecting hoses.

Light oil or soapy water
Loctite 567 or equivalent

1. Cut 2 pieces of 1/2 inch tubing 20 feet (6.1 m) long.
2. Lubricate the barb fitting and/or hose I.D. with a light oil or soapy water only. Do not use heavy oil or grease.

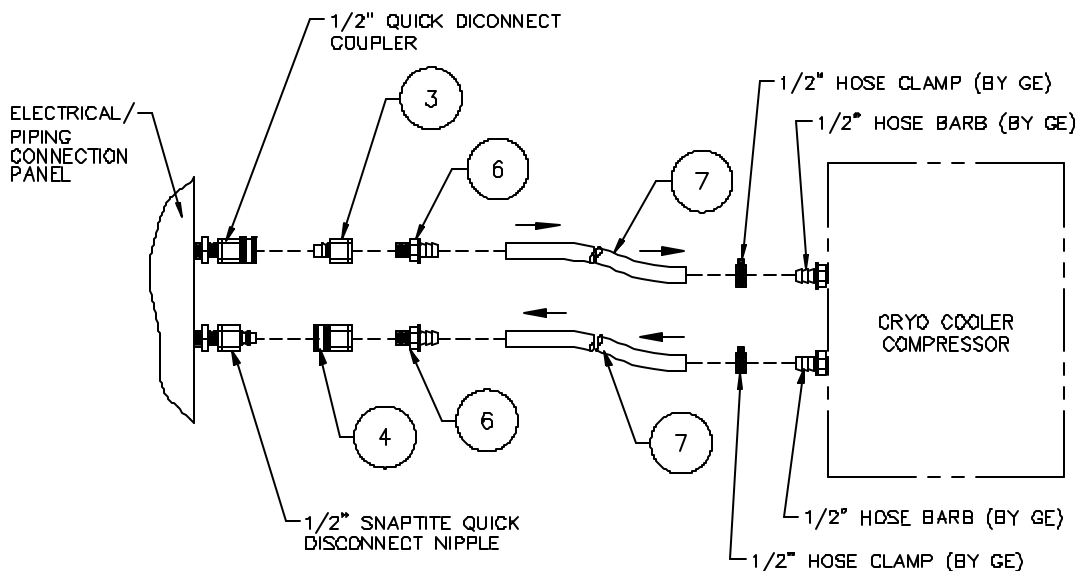


Illustration 17 HOSE TO HOSE BARB ASSEMBLY

Refer to table 12, item numbers 3 to 7

3. Insert the fitting into the hose until the barb is in the hose.
4. Place the end of the fitting against a wood block. Grip the hose firmly at approximately one inch (2.54 cm) from the end of the hose. Push with steady force until the end of the hose bottoms on the fitting and is covered by the yellow plastic cap.
5. Apply a generous amount of LocTite 567 or equivalent to the male pipe threads of each hose barb fitting.

6. Assemble and mark each hose per table.

HOSE SECTION Feet (m)	QUICK DISCONNECT	MARK HOSE AS:
60	Male	Cryo cooler coolant from TSCC
60	Female	Cryo cooler coolant to TSCC

Table 15 **Hose And Quick Disconnect**

7. Tighten each threaded connection to prevent leaks.
8. Using hose clamps (installer supplied) attach each hose assembly to appropriate connection of cryo cooler.

5-1-3 Condenser Water Supply Installation

The following items are provided by the installer and are required for the proper installation of the condenser water lines.

Loctite 567 or equivalent

NOTE 1: The installer should use flexible line to short loop the condenser water supply and return lines and purge for several hours before connecting the condenser water supply to the TSCC. Any debris from installing the copper lines could cause blockage in the brazed-plate heat exchanger, even if the water is normally particulate free, newly installed piping needs to be flushed out.

1. Locate two 1 ½" FPT fittings on left rear of the TSCC.
2. Apply a generous amount of Loctite 567 or equivalent to the male pipe threads of the supply and return pipes.
3. Connect the supply line to the 1 ½" fitting on the right, labeled CONDENSER WATER IN. See Illustration 18. Tighten the connection snugly to prevent leaks.
4. Connect the return line to the 1 ½" fitting on the left, labeled CONDENSER WATER OUT. See Illustration 18. Tighten the connection snugly to prevent leaks.

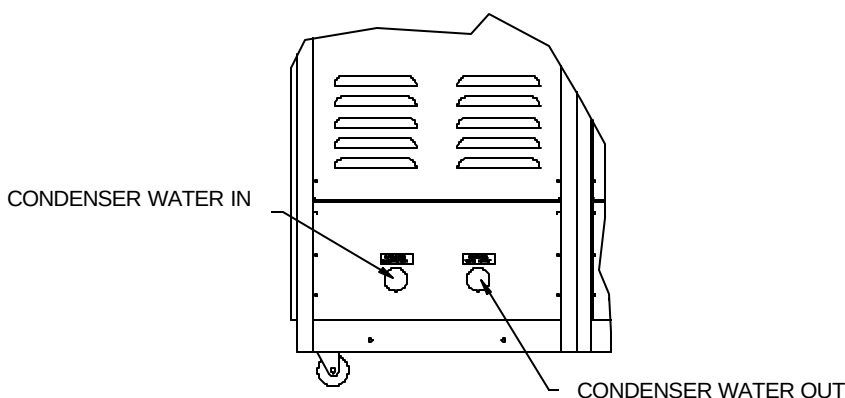


Illustration 18 **CONDENSER WATER PIPING**

6 SYSTEM OPERATION

The following procedures are intended to give instruction on the TSCC system controls and routine operation. Initial start-ups after installation, will be performed by GE personnel and system operation will be verified after installation by an Ellis and Watts approved service provider. Start-ups after repairs should be conducted by the customers service provider. It is assumed at this point, that the TSCC and associated equipment have gone through initial startup previously and is ready for operation. If the unit has stopped due to a failure, refer to section 6-1-4.

6-1 OPERATOR START-UP PROCEDURE

6-1-1 Starting Up The TSCC and Associated Equipment

1. Verify that all fluid lines are connected and not leaking. The fluid lines are located on the rear of the unit.
2. Check the coolant level in the reservoir. Verify that the coolant level is between the add and full labels. The reservoir sight glass is located on the top shelf inside the right panel (See Illustration 19). If it is not refer to Section 6-1-4.

NOTE: If the unit is being started up for the first time the reservoir needs to be continuously filled during the start-up operation to purge the system of air.

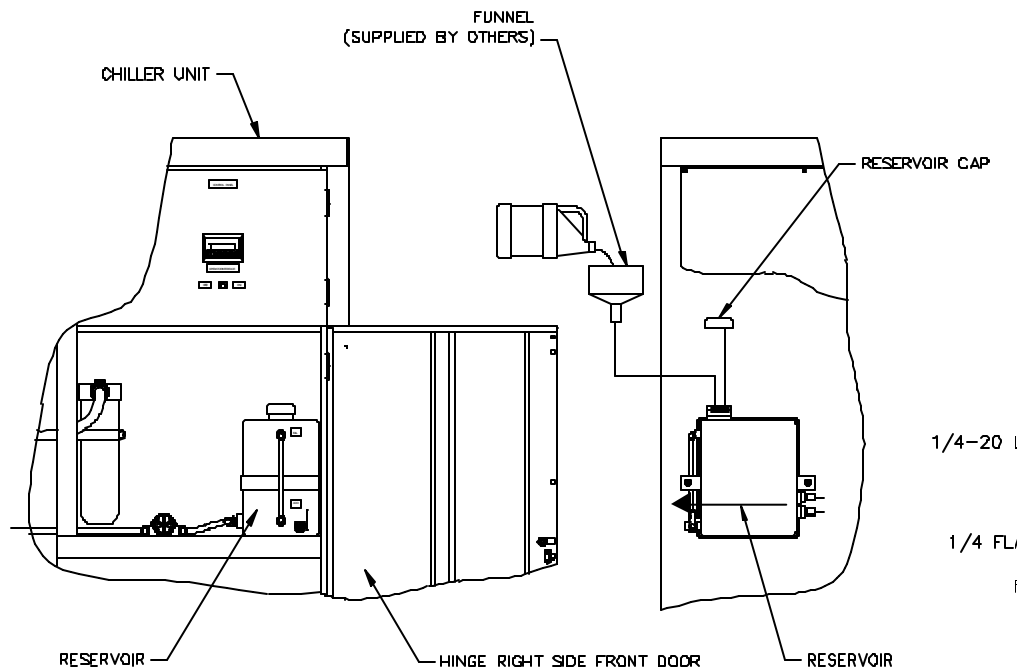


Illustration 19 **RESERVOIR SIGHT GLASS**

CAUTION

During periods where compressor crankcase heaters are without power, liquid refrigerant can accumulate in compressor crankcase. Restarting compressor without allowing sufficient time for crankcase heaters to vaporize liquid refrigerant can cause serious internal damage to compressor.

3. The Control Power Switch is located on the top front panel (see Illustration 19). Verify toggle switch is switched off. If the main disconnect switch is OFF, set to ON by rotating red handle clockwise until handle is in vertical position. The main disconnect switch is located on the back lower right side of the unit (see Illustration 15). **(Wait a minimum of two hours before continuing.)**
4. Verify that the disconnects for the Cryo and Gradient Cooler are in the "ON" position. If not, set to ON as with the TSCC Main Disconnect procedure in Step 3, (see Illustration 15 For locations).
5. Place the Control Power Switch to the "ON" Position. TSCC should start after 5 seconds and be fully functional after 30 seconds. If the TSCC shuts down after 30 seconds, refer to section 6-1-4.

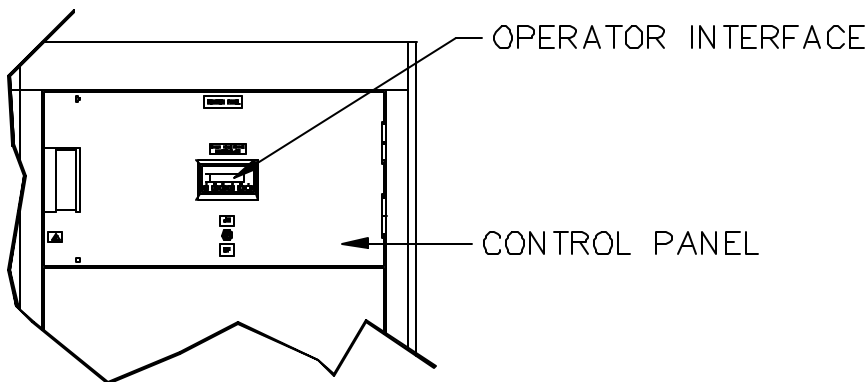


Illustration 20 **CONTROL PANEL**

6. After the initial 30 seconds of operation, set power switches to "ON" for the Gradient and Cryo Cooler.
7. After 30 minutes of operation verify that the unit is maintaining proper temperatures and flows as shown in table 13.

NOTE: Two globe valves are located on the coolant supply lines (gradient and cryo loops) for flow adjustment. The valves have been set for 60 Hz operation. For 50 Hz operation, the valves should be adjusted to obtain the flow shown in Table 16.

	Coolant Flow	Coolant Temperature
Gradient	6.1 GPM + 0.5 GPM	20° +/- 1C
Cryo Cooler Flow	2.1 GPM + 0.5 GPM	20° +/- 1C

Table 16 **TSCC Temperature And Flow Tolerances**

6-1-2 Shutting Down The TSCC

1. Switch to "OFF" the Gradient and Cryo Coolers.
2. Verify that the red led on the Operator Interface is not on. If the led is glowing, check the alarm section of the Operator Interface for any faults and log them before proceeding. Refer to section 6-1-4 for further instructions.
3. Set the Control Power Switch to OFF.

6-1-3 Maintenance

An approved service provider should conduct routine maintenance every 4 months. This maintenance consists of:

Verifying proper flows and temperatures per table 13 and logging data.

Change coolant filter.

Leak test refrigerant system.

Check electrical control panel for burned or frayed wiring and loose connections.

Checking all coolant hoses for obvious leaks and for worn areas.

Verifying proper operation of refrigeration system.

Check condenser coils for dirt or damage.

Contact GE for current schedule or approved service provider.

6-1-4 General Troubleshooting

The TSCC chiller system is designed to be monitored by the GE Service Center at all times. Some occasions arise where the operator can assist in troubleshooting. In all cases the operator will be **most useful** by checking for and logging any faults on the Operator Interface and noting any abnormal conditions in the equipment room where the TSCC is located. Refer to sections 6-2-1 through 6-2-3 for instruction on how to locate faults on Operator Interface screens.

6-2 PLC OPERATION

6-2-1 Operator Interface Display Description:

The operator interface is located on the front panel of the TSCC and displays the model number of the unit, the software revision of the programmed PLC, parameter set points, parameter data, parameter alarm status, and system run time. The interface also allows the user to change the operating set points for the fluid coolant circuit. The upper and lower alarm limits for the various parameters are preset by the manufacturer and are not accessible by service. See Illustration 20 for an example of the layout of the display, user buttons, power switch and indicator lights.

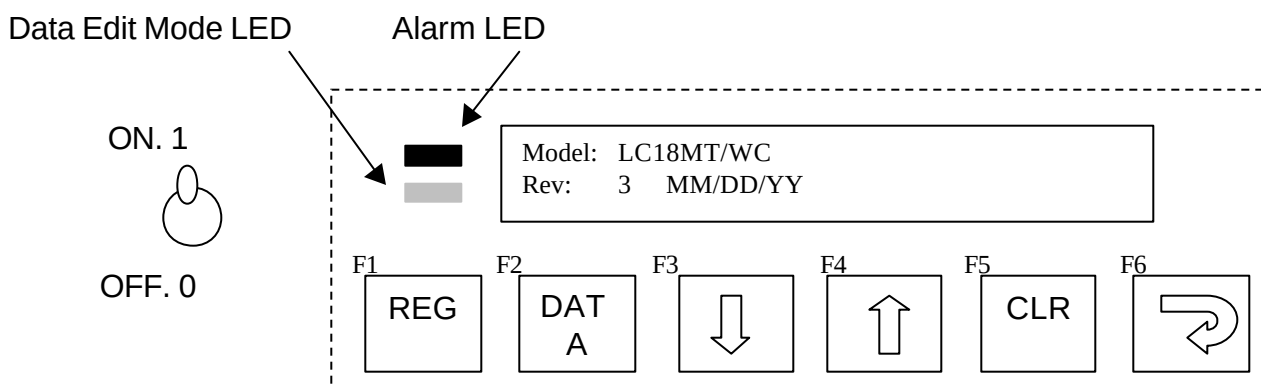


Illustration 21 **OPERATOR INTERFACE**

6-2-2 Button Description

- Button F1 Toggles the display between Menu 1 and Menu 2.
- Button F2 Puts the unit into data edit mode for changing the operating set point for supply coolant temperature. The green led will light when the unit is in data edit mode.
- Button F3 Menu List Display: Steps through the menu list in the forward direction.
Change Set Point: Increases the value shown on the display in 0.1 unit increments.
- Button F4 Menu List Display: Steps through the menu list in the reverse direction.
Change Set Point: Decreases the value shown on the display in 0.1 unit increments.
- Button F5 To extinguish the red alarm status indicator. Note: The operating parameter must be in a normal or no fault mode in order to turn off the red alarm status indicator.
- Button F6 Stores new set point changes and exits the data edit mode.

6-2-3 Menu Lists

1. Press the F1 Button to change between Menu 1 and Menu 2.
2. Refer to Table 23, Menu 1 – System Parameter Data, to view the listing of the following types of items: Model Number and Software revision of the unit, operating set points for specific parameters, current parameter data measurement, and total run hours. The first line item on Menu 1 will show the Model Number of the Unit.
3. Refer to Table 24, Menu 2 – System Operating Status, to view the listing of the alarm or failure conditions. The first line item on Menu 2 will show “Supply Fluid Temperature”.

6-2-4 Changing The Set Points

The coolant temperature set points are preset at Ellis and Watts per G.E. specifications. There is a 2°C allowable variance, settable by the operator. If there is a need to change the set point value, then refer to the following steps to perform the change.

1. Press F2 to enter the data edit mode.
2. Press either the F3 or F4 buttons to change the set point value.
3. Press F6 to store the set point values and exit the data edit mode.

Menu 1 – System Parameter Data

Example of Model Number and Revision	
Model Number:	LC18MT/WC
Rev.: #	Mo/Dy/Yr
Parameter	Range Value / Unit of Measure (see NOTE 1)
Supply Temp:	0 to 29.4 Degrees Centigrade
Return Temp:	0 to 50 Degrees Centigrade
Temp Set Point:	17 to 24 Degrees Centigrade (Factory set 20°C)
Gradient Flow:	0 to 6.0 GPM +/- 10%
Cryo-Cooler Flow:	0 to 2.5 GPM +/- 10%
TSCC 3 way valve % open:	0 to 100%
Total Run Hours:	0 to 9999 Hours NOTE 2
NOTE 1: Unit of measure is set at the factory.	
NOTE 2: After the run hours reaches 9999, the unit resets to zero.	

Table 17 **System Parameter Data**

Menu 2 – System Operating Status

Parameter	Display of Operating Mode		
Supply Temp:	Low	Normal	High
Coolant Level:	Low	Normal	
Gradient Flow:	Failure	Low	Normal
Cryo-Cooler Flow:	Failure	Low	Normal
Suction Pressure	Fault	No Fault	
Discharge Pressure	Fault	No Fault	
Compressor / Phase	Fault	No Fault	
Compressor Cycling:	Normal	- - - - (See NOTE 1)	
NOTE 1: Each dashed line represents an attempt by the PLC to restart the chiller. After 5 attempts the TSCC will shut down.			

Table 18 **System Operating Status**

TERMS & ACRONYMS

GEMS	General Electric Medical Systems
MDP	Main Distribution Panel
QD	Quick disconnect coupler
RGA	Returned Goods Authorization
EPA	Environmental Protection Agency
MRI	Magnetic Resonance Imaging
UBC	United Builders Code
SST	Stainless Steel
NPTF	National Pipe Thread, Female
DB	Distribution Block
PLC	Programmable Logic Control
TSCC	Twin System Cooling Cabinet
Liquidtite	Trade name for water tight flexible conduit
LocTite	Brand name for adhesives, sealants, and threadlockers

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MATERIAL SAFETED DATA SHEETS

Forane ® R - 134a
Tetrafluoroethane (HFC - 134a)

Dowfrost ® Heat Transfer Fluid
Propylene glycol
Dipotassium Phosphate
Deionized Water

The Dow Chemical Company

Material Safety Data Sheet

Midland, Michigan 48674

Emergency 517 - 636-4400

1. CHEMICAL PRODUCT & COMPANY IDENTIFICATION Page: 1

24-Hour Emergency Phone Number: 989-636-4400

Product: DOWFROST* HEAT TRANSFER FLUID

Product Code: 23545

Effective Date: 01/08/99 Date Printed: 05/07/01 MSD: 000130

The Dow Chemical Company, Midland, MI 48674

Customer Information Center: 800-258-2436

2. COMPOSITION/INFORMATION ON INGREDIENTS

Propylene glycol	CAS# 000057-55-6	95%
Dipotassium phosphate	CAS# 007758-11-4	<5%
Deionized water	CAS# 007732-18-5	<5%

3. HAZARDS IDENTIFICATION

EMERGENCY OVERVIEW

* Colorless, odorless liquid. Toxic fumes released in fire * situations.

POTENTIAL HEALTH EFFECTS (See Section 11 for toxicological data.)

EYE: May cause slight transient eye irritation. Corneal injury is unlikely.

SKIN: Essentially nonirritating to skin on prolonged contact. A single prolonged skin exposure is not likely to result in the material being absorbed through skin in harmful amounts. Repeated exposures may cause slight flaking, tenderness and softening of skin.

(Continued on page 2)

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MATERIAL SAFETY DATA SHEET PAGE: 2

Product: DOWFROST* HEAT TRANSFER FLUID Product Code: **23545**Effective Date: 01/08/99 Date Printed: 05/07/01 MSD: **000130**

INGESTION: Single dose oral toxicity is low. No hazards anticipated from swallowing small amounts incidental to normal handling operations.

INHALATION: A single prolonged (hours) inhalation exposure is not likely to cause adverse effects. Mists are not likely to be hazardous.

SYSTEMIC (OTHER TARGET ORGAN) EFFECTS: Repeated excessive ingestion may cause central nervous system effects.

CANCER INFORMATION: Did not cause cancer in long-term animal studies.

TERATOLOGY (BIRTH DEFECTS): Birth defects are unlikely. Exposures having no adverse effects on the mother should have no effect on the fetus.

REPRODUCTIVE EFFECTS: In animal studies, has been shown not to interfere with reproduction.

4. FIRST AID

EYE: Flush eyes with plenty of water.

SKIN: Wash off in flowing water or shower.

INGESTION: If swallowed, seek medical attention. Do not induce vomiting unless directed to do so by medical personnel.

INHALATION: Remove to fresh air if effects occur. Consult a physician.

NOTE TO PHYSICIAN: No specific antidote. Supportive care. Treatment based on judgment of the physician in response to reactions of the patient.

5. FIRE FIGHTING MEASURES

FLASH POINT: 215F, 102C METHOD USED: TOC (Tag Open Cup)

(Continued on page 3)

* OR (R) INDICATES A TRADEMARK OF THE DOW CHEMICAL COMPANY

MATERIAL SAFETY DATASHEET PAGE: 3

Product: DOWFROST* HEAT TRANSFER FLUID

Product Code: 23545

Effective Date: **01/08/99** Date Printed: 05/07/01 MSD: 000130-----
FLAMMABLE LIMITS LFL: 2.6% @ 100C UFL: 12.5% @ 130C

HAZARDOUS COMBUSTION PRODUCTS: During a fire, smoke may contain the original material in addition to unidentified toxic and/or irritating compounds. Hazardous combustion products may include and are not limited to carbon monoxide, carbon dioxide.

OTHER FLAMMABILITY INFORMATION: Violent steam generation or eruption may occur upon application of direct water stream. Flammable concentration of vapor can accumulate at temperatures above 215.0 deg. F. Liquid mist of this product can burn. Spills of these organic liquids on hot fibrous insulations may lead to lowering of the autoignition temperatures possibly resulting in spontaneous combustion.

EXTINGUISHING MEDIA: Water fog or fine spray, carbon dioxide, dry chemical, foam. Alcohol resistant foams (ATC type) are preferred if available. General purpose synthetic foams (including AFFF) or protein foams may function, but much less effectively. Do not use direct water stream. Will spread fire.

MEDIA TO BE AVOIDED: Do not use direct water stream.

FIRE FIGHTING INSTRUCTIONS: Keep people away. Isolate fire area and deny unnecessary entry. Burning liquids may be moved by flushing with water to protect personnel and minimize property damage. Burning liquids may be extinguished by dilution with water. Do not use direct water stream. May spread fire.

PROTECTIVE EQUIPMENT FOR FIRE FIGHTERS: Wear positive-pressure self-contained breathing apparatus (SCBA) and protective firefighting clothing (includes fire-fighting helmet, coat, pants, boots, and gloves). If protective equipment is not available or not used, fight fire from a protected location or safe distance.

6. ACCIDENTAL RELEASE MEASURES (See Section 15 for Regulatory Information)

(Continued on page 4)

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MATERIAL SAFETY DATA SHEET PAGE: 4

Product: DOWFROST* HEAT TRANSFER FLUID

Product Code: 23545

Effective Date: 01/08/99 Date Printed: 05/07/01 MSD: 000130

PROTECT PEOPLE: Clear non-emergency personnel from area.

PROTECT THE ENVIRONMENT: Contain liquid to prevent contamination of soil, surface water or ground water.

CLEANUP: Clean up with absorbent material. Sweep up.

7. HANDLING AND STORAGE

HANDLING: Product on surfaces can cause slippery conditions.

STORAGE: Store below 121 C, 250 F.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

ENGINEERING CONTROLS: Good general ventilation should be sufficient for most conditions.

PERSONAL PROTECTIVE EQUIPMENT

EYE/FACE PROTECTION: Use safety glasses.

SKIN PROTECTION: For brief contact, no precautions other than clean body-covering clothing should be needed. Use impervious gloves when prolonged or frequently repeated contact could occur.

RESPIRATORY PROTECTION: No respiratory protection should be needed.

EXPOSURE GUIDELINE(S): Propylene glycol: AIHA WEEL is 50 ppm total, 10 mg/m³ aerosol only.**9. PHYSICAL AND CHEMICAL PROPERTIES**

APPEARANCE: Colorless. ODOR: Odorless liquid. VAPOR PRESSURE: 0.22 mmHg @ 20C, 68F VAPOR DENSITY: 2.62 BOILING POINT: 370F, 188C SOLUBILITY IN WATER: Complete

(Continued on page 5)

* OR (R) INDICATES A TRADEMARK OF THE DOW CHEMICAL COMPANY

M A T E R I A L S A F E T Y D A T A S H E E T **PAGE: 5**

Product: DOWFROST* HEAT TRANSFER FLUID

Product Code: 23545

Effective Date: 01/08/99 Date Printed: 05/07/01 MSD: 000130

SPECIFIC GRAVITY: 1.050 @ 60/60F, 16C

10. STABILITY AND REACTIVITY

CHEMICAL STABILITY: Thermally stable at typical use temperatures.

CONDITIONS TO AVOID: Avoid temperatures above 121C/250F. Product can decompose at elevated temperatures.

INCOMPATIBILITY WITH OTHER MATERIALS: Avoid contact with oxidizing materials.

HAZARDOUS DECOMPOSITION PRODUCTS: Hazardous decomposition products depend upon temperature, air supply and the presence of other materials.

HAZARDOUS POLYMERIZATION: Will not occur.

11. TOXICOLOGICAL INFORMATION (See Section 3 for Potential Health Effects. For detailed toxicological data, write or call the address or non-emergency number shown in Section 1)

SKIN: The LD50 for skin absorption in rabbits is greater than 10,000 mg/kg.

INGESTION: The oral LD50 for female rats is about 20.3 g/kg.

MUTAGENICITY (EFFECTS ON GENETIC MATERIAL): Results of in vitro (test tube) mutagenicity tests have been negative. Results of mutagenicity tests in animals have been negative.

12. ECOLOGICAL INFORMATION (For detailed Ecological data, write or call the address or non-emergency number shown in Section 1)

ENVIRONMENTAL FATE MOVEMENT AND PARTITIONING: Based largely or completely on data for major component(s). Bioconcentration potential is low (BCF less than 100 or Log Pow less than 3).

DEGRADATION AND PERSISTENCE: Based largely or completely on data for major component(s). Biodegradation under aerobic static laboratory conditions is high (BOD20 or BOD28/ThOD

(Continued on page 6)

* OR (R) INDICATES A TRADEMARK OF THE DOW CHEMICAL COMPANY

MATERIALSAFETY DATASHEET PAGE: 6

Product: DOWFROST* HEAT TRANSFER FLUID

Product Code: 23545

Effective Date: 01/08/99 Date Printed: 05/07/01 MSD: 000130

greater than 40%).

ECOTOXICITY: Based largely or completely on data for major component(s). Material is practically non-toxic to aquatic organisms on an acute basis (LC50 greater than 100 mg/L in most sensitive species).

13. DISPOSAL CONSIDERATIONS (See Section 15 for Regulatory Information)

DISPOSAL: DO NOT DUMP INTO ANY SEWERS, ON THE GROUND OR INTO ANY BODY OF WATER. All disposal methods must be in compliance with all Federal, State/Provincial and local laws and regulations. Regulations may vary in different locations. Waste characterizations and compliance with applicable laws are the responsibility solely of the waste generator. THE DOW CHEMICAL COMPANY HAS NO CONTROL OVER THE MANAGEMENT PRACTICES OR MANUFACTURING PROCESSES OF PARTIES HANDLING OR USING THIS MATERIAL. THE INFORMATION PRESENTED HERE PERTAINS ONLY TO THE PRODUCT AS SHIPPED IN ITS INTENDED CONDITION AS DESCRIBED IN MSDS SECTION 2 (Composition/information On Ingredients).

FOR UNUSED & UNCONTAMINATED PRODUCT, the preferred options include sending to a licensed, permitted: recycler, reclaimer, or waste water treatment system.

As a service to its customers, Dow can provide names of information resources to help identify waste management companies and other facilities which recycle, reprocess or manage chemicals or plastics, and that manage used drums. Telephone Dow's Customer Information Center at 800-258-2436 or 517-832-1556 for further details.

14. TRANSPORT INFORMATION

CANADIAN TDG INFORMATION: For TDG regulatory information, if required, consult transportation regulations, product shipping papers, or your Dow representative.

15. REGULATORY INFORMATION (Not meant to be all-inclusive- -selected regulations represented)

NOTICE: The information herein is presented in good faith and

(Continued on page 7)

* OR (R) INDICATES A TRADEMARK OF THE DOW CHEMICAL COMPANY

MATERIALSAFETY 0ATASHEET PAGE: 7

Product: DOWFROST* HEAT TRANSFER FLUID

Product Code: 23545

Effective Date: **01/08/99** Date Printed: 05/07/01 MSD: 000130

believed to be accurate as of the effective date shown above. However, no warranty, express or implied is given. Regulatory requirements are subject to change and may differ from one location to another; it is the buyer's responsibility to ensure that its activities comply with federal, state or provincial, and local laws. The following specific information is made for the purpose of complying with numerous federal, state or provincial, and local laws and regulations. See other sections for health and safety information.

U.S. REGULATIONS

SARA 313 INFORMATION: To the best of our knowledge, this product contains no chemical subject to SARA Title III Section 313 supplier notification requirements.

SARA HAZARD CATEGORY: This product has been reviewed according to the

EPA "Hazard Categories" promulgated under Sections 311 and 312 of the Superfund Amendment and Reauthorization Act of **1986** (SARA Title 111) and is considered, under applicable definitions, to meet the following categories:

Not to have met any hazard category

TOXIC SUBSTANCES CONTROL ACT (TSCA):

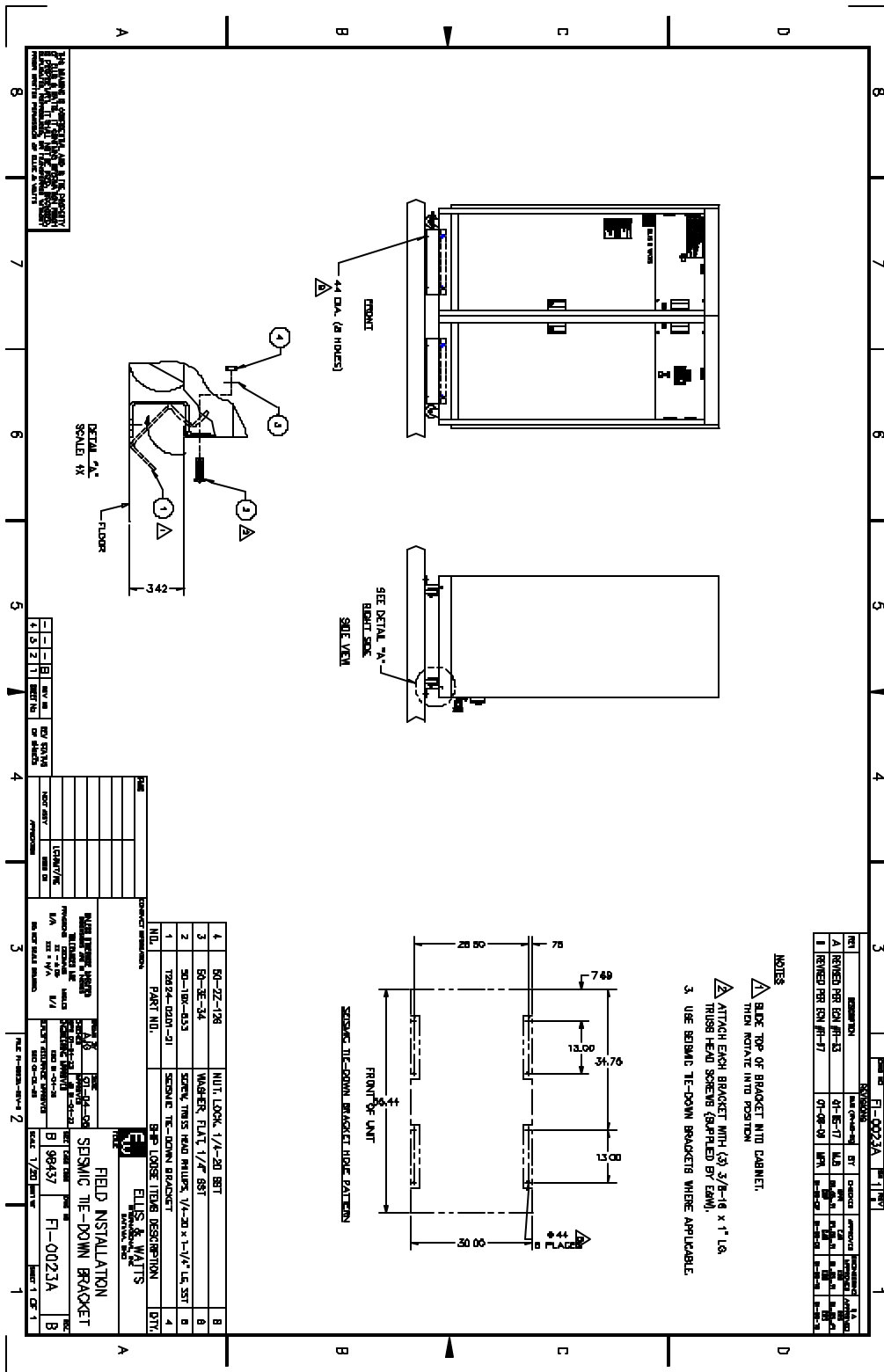
All ingredients are on the TSCA inventory or are not required to be listed on the TSCA inventory.

STATE RIGHT-TO-KNOW: The following product components are cited on certain state lists as mentioned. Non-listed components may be shown in the composition section of the MSDS.

(Continued on page 8)

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FI DRAWINGS



Ellis & Watts International, Inc.
Batavia, Ohio 45103

